

Lesson 6: Problem Solving with Triangles

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Lesson 6: Problem Solving with Triangles

The Pythagorean Theorem ✓

Find Missing Angles ✓

Missing Sides in Similar Figures ✓

Indirect Measurement ✓

The Pythagorean Theorem

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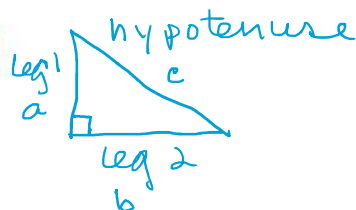
THE PYTHAGOREAN THEOREM

A right triangle has a right angle, which equals 90° . The two sides that form the right angle are called **legs**, and the side opposite the right angle is the **hypotenuse**. The hypotenuse is always the longest side of a right triangle.

The relationship between the legs and hypotenuse of a right triangle was discovered almost 2,500 years ago by a Greek mathematician named Pythagoras.

Pythagorean theorem $a^2 + b^2 = c^2$

The sum of the squares of the legs of a right triangle equals the square of the hypotenuse.



$$(Leg 1)^2 + (Leg 2)^2 = (Hypotenuse)^2$$

$$a^2 + b^2 = c^2$$

$$c = \sqrt{a^2 + b^2}$$



$$= \sqrt{81 + 144}$$

$$12 \quad c = \sqrt{225}$$

$$c = 15$$

FINDING THE HYPOTENUSE

Example The lengths of the legs of a right triangle are 9 and 12. What is the hypotenuse?

Step 1

Substitute 9 for a and 12 for b in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$

$$9^2 + 12^2 = c^2$$

The hypotenuse is 15 units long.

Step 2

Evaluate the left side of the equation.

$$\sqrt{81 + 144} = c$$

$$\sqrt{225} = c$$

Step 3

Find c .

$$15 = c$$

$$c = 15$$

BAD FORM

FINDING THE LEG OF A RIGHT TRIANGLE

Example The hypotenuse of a right triangle has a length of 10 units, and one of the legs is 6 units long. Find the measure of the other leg.

Step 1

Substitute 10 for c and 6 for a in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$

$$6^2 + b^2 = 10^2$$

The other leg is 8 units long.

Step 2

Simplify the equation.

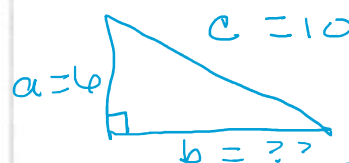
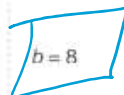
$$36 + b^2 = 100$$

$$b = \sqrt{64}$$

Step 3

Find b .

$$b = 8$$



$$a^2 + b^2 = c^2$$

$$b^2 = c^2 - a^2$$

$$b = \sqrt{c^2 - a^2}$$

$$= \sqrt{10^2 - 6^2}$$

$$= \sqrt{64}$$

$$b = 8$$

Pythagorean Theorem - Examples Set #1

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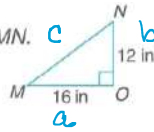
1. Find the measurement of side MN.

$$c = \sqrt{12^2 + 16^2}$$

$$= 144 + 256$$

$$= 400$$

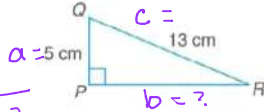
$$c = 20 \text{ m}$$



2. What is the measurement of side PR?

$$b = \sqrt{13^2 - 5^2}$$

$$b = 12 \text{ cm}$$



3. The legs of a right triangle measure 10 inches and 24 inches. What is the hypotenuse?

$$c = \sqrt{10^2 + 24^2}, c = \sqrt{676}, c = 26 \text{ in}$$

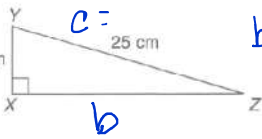
4. What is the measurement of side XZ?

$$b = \sqrt{c^2 - a^2}$$

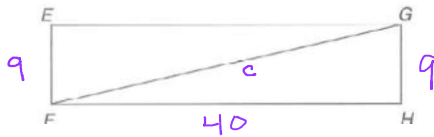
$$b = \sqrt{25^2 - 7^2}$$

$$b = \sqrt{576}$$

$$b = 24 \text{ cm}$$



5. In the rectangle, side EF is 9 cm and side FH is 40 cm. Find the measure of the diagonal FG.



$$c = \sqrt{40^2 + 9^2}$$

$$c = 41$$

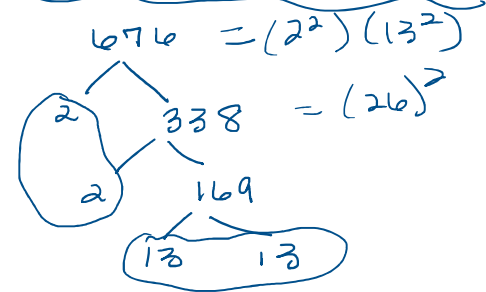
$$① a^2 + b^2 = c^2$$

$$② c = \sqrt{a^2 + b^2}$$

$$③ a = \sqrt{c^2 - b^2}$$

$$④ b = \sqrt{c^2 - a^2}$$

PRIME FACTORIZATION TREE



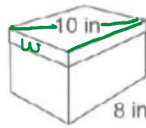
9, 40, 41 is a PYTHAGOREAN TRIPLET

7, 24, 25 } also
10, 24, 26 }

Pythagorean Theorem - Examples Set #2

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6. If the lid of a box has a diagonal length of 10 inches and a length of 8 inches, what is the width of the box in inches?



$$L^2 + w^2 = \text{Diagonal}^2$$

$$8^2 + w^2 = 10^2$$

$$w = \sqrt{100 - 64}$$

$$\boxed{w = 6 \text{ in}}$$

7. For a circus act, an 18-foot pole is braced with a wire that extends from the top of the pole to a stake in the ground 80 feet from the base of the pole. How long is the bracing wire?

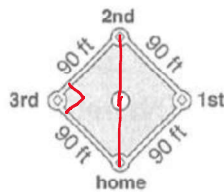


$$c = \sqrt{18^2 + 80^2}$$

$$= \sqrt{324 + 6400}$$

$$\boxed{c = 82 \text{ ft}}$$

8. The base paths on a baseball diamond form right angles. To the nearest tenth of a foot, what is the distance from home plate to second base?



$$c = \sqrt{90^2 + 90^2}$$

$$= \sqrt{2(8100)}$$

$$= \sqrt{16200}$$

$$c = 127.279 \text{ ft.}$$

your instructor
thought home plate
was the center!!

~~$$a^2 + b^2 = c^2$$

$$a = b, c = 90$$

$$a^2 + a^2 = 90^2$$

$$2a^2 = 90^2$$

$$a^2 = 8100/2$$

$$a^2 = 4050$$

$$a = \sqrt{4050}$$

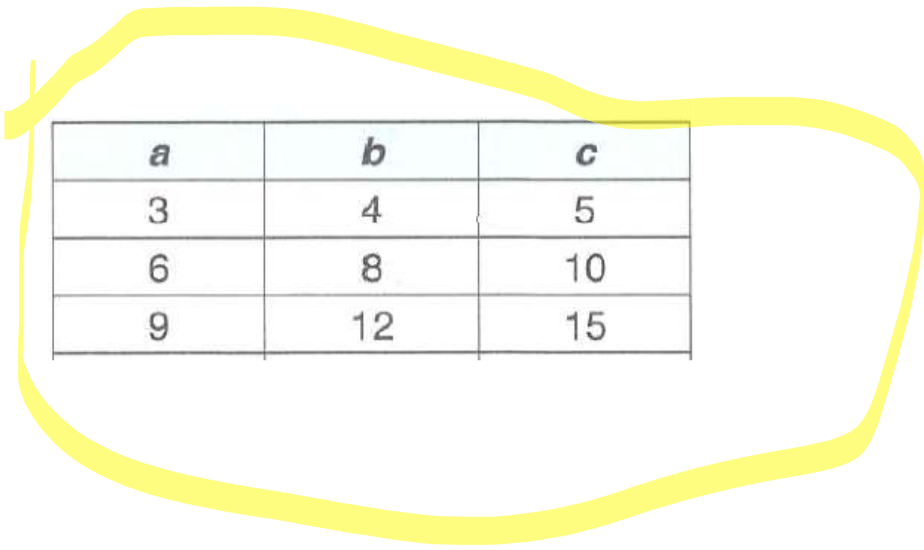
$$a = 63.6396 \dots$$~~

Summary

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A set of whole numbers that forms a right triangle is called a **Pythagorean triple**.



<i>a</i>	<i>b</i>	<i>c</i>
3	4	5
6	8	10
9	12	15

Finding Missing Angles

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This is a repeat of what we learned in Lesson 1, Angles.